

Java Program – Marks

1. Objective

To develop a Java program that accepts marks of five subjects, calculates the total marks and average marks using a loop, and displays the results.

2. Task Description

Write a Java program using the Scanner class to read five marks from the user. Use a for loop to repeatedly accept the marks and add them to a total. Calculate the average by dividing the total by 5.0 and display both total and average marks.

3. Step-wise Execution

1. Start the program.
2. Import the Scanner class.
3. Create a Scanner object to read input from the keyboard.
4. Initialize total to 0.
5. Display a message asking the user to enter 5 marks.
6. Run a for loop from 1 to 5.
7. Read each mark using `sc.nextInt()`.
8. Add each mark to total.
9. Calculate average using `total / 5.0`.
10. Display the total marks.
11. Display the average marks.
12. Close the Scanner object.
13. Stop the program.

4. Algorithm

```
Step 1: Start.
Step 2: Create Scanner object sc.
Step 3: Set total = 0.
Step 4: Repeat the following steps for i = 1 to 5:
    a) Read marks.
    b) Add marks to total.
Step 5: Calculate average = total / 5.0.
Step 6: Print total marks.
Step 7: Print average marks.
Step 8: Close Scanner.
Step 9: Stop.
```

5. UML Diagram

Marks	
Attributes	total : int
Methods	main(String[] args) : void
Input	5 subject marks

Output	Total Marks, Average Marks
--------	----------------------------

6. Program Code

```
package core_java;

import java.util.Scanner;

public class Marks {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int total = 0;

        System.out.println("Enter 5 marks:");

        for (int i = 1; i <= 5; i++) {
            int marks = sc.nextInt();
            total = total + marks;
        }

        double average = total / 5.0;

        System.out.println("Total Marks = " + total);
        System.out.println("Average Marks = " + average);

        sc.close();
    }
}
```

7. Output Screen

```
Enter 5 marks:
80
75
90
85
70
Total Marks = 400
Average Marks = 80.0
```

8. Result

The Java program was executed successfully. It accepts five marks and correctly calculates and displays the total and average marks.